Central Maintenance Computer – General data

1. General

The Central Maintenance Computer (CMC) is used to:

- Verify the configuration of the Apex system
- Determine the status of an LRU or function
- Execute supported LRU or function test capabilities
- Diagnose a current Apex problem
- Use historical data to diagnose past Apex problems
- Download the CMC fault history database
- Verify the configuration of the CMC

A. Central Maintenance Computer Function (CMCF)

A software application hosted on the MAU advanced graphics module (AGM1). This software provides the real-time collection of member system fault reports and the subsequent storage of maintenance messages and any associated correlations to flight deck effects. Additionally, this application provides the maintenance system central point of control on the aircraft.

The Apex CMC is constrained to provide maintenance support for only the Honeywell Apex Avionics System. It does not provide maintenance capability for the other aircraft systems (for example, engine, hydraulics, etc.).

B. Loadable Diagnostic Interface (LDI)

A database that contains specific information pertaining to the maintenance characteristics of the Aircraft Diagnostics and Maintenance System (ADMS). The LDI provides the necessary operational data to permit the generic CMCF to operate within the particular aircraft.

C. Fault History Database (FHDB)

The storage repository for all fault data that is output from the CMCF. This is the data that is accessed by the maintenance technician, using the remote terminal, to assess the past and present operating condition of the aircraft.

D. Remote Terminal (RT)

A PC-based application, consisting of a PC, an operating system, and an ADMS application. This system connects to the CMCF via the aircraft local area network (LAN). The RT provides the means by which the user gains access to the aircraft-resident CMCF. The primary function of the RT is to provide the user interface, through which the capabilities provided by the ADMS are selected and controlled.

E. Software Loading

The CMC Remote Terminal data loading system will automatically load the software and data bases on the appropriate MAU module(s) if full load is selected.

Note Depending if Aircraft has Honeywell Apex Phase 1A or Apex Phase 2A software installed, the software required is listed in Table 1 — MAU Module Software/Database Locations – Phase 1A Load 1.3 or Table 2 — MAU Module Software/Database Locations – Phase 2A Load 4.4.

1) Specific MAU Module – Software Loading Procedures

For an individual MAU module replacement, the correct software loading sequence must be followed.

Note For ECL load procedures refer to 46-20-01, Apex Electronic Checklist File – Load Procedures.

2) Multiple MAU Modules – Software Loading Procedures

For multiple or complete MAU modules replacement, the correct software loading sequence must be followed.

3) The MAU Module Software Loading sequence is as follows:

a) APEX MAU Software via the 'CMC Remote Terminal' (refer to 45-00-02, Apex Modular Avionics Unit Software – Load Procedures). Media CD to be used PN: xx60000469–xxx – Apex 12 Load 1.3 (Pre Mod 6/2303); or PN: xx60002735–xxx – Apex 12 Load 4.4 (Post Mod 6/2303).

b) Load the APM Settings File via the 'CMC Remote Terminal' (refer to 45-00-01, Apex Options and Settings File – Load Procedures). Media CD to be used PN: VSP25–xxxxx.

c) Load the APM Options File via the 'CMC Remote Terminal' (refer to 45-00-01, Apex Options and Settings File – Load Procedures). Media CD to be used PN: xx60000053–xxxxx (Pre Mod 6/2303); or PN: xx600002657–xxxxx (Post Mod 6/2303) – Options File password required (on CD).

Note Post Mod 6/2303 aircraft with S.O.O. 6221 Automatic Flight Control System (Autopilot) incorporated have two Option Files, one CD for autopilot activated and the other for autopilot deactivated. Ensure the correct Options File is loaded.

d) Load the MAU LDI (Loadable Diagnostics Interface) via 'CMC Remote Terminal'. Media CD to be used PN: VD6–LDI_1_1_20 or higher version (Pre Mod 6/2303); or PN: VD6_LDI_4_1_4_20141017 or higher version (Post Mod 6/2303).

e) FMS Performance Database via the 'CMC Remote Terminal' (refer to 45-00-03, Apex FMS Aircraft Performance Database Files – Load Procedures). Media CD to be used PN: xx60000461–xxx – APM FMS Performance Database.

f) All navigation database CD's (NAV Database (Purple), Charts Database (Blue), Terrain Database (Green), etc) via the 'CMC Remote Terminal' (refer to 45-00-04, Apex Database Files – Load Procedures). No special loading sequence is required between databases.

g) Electronic checklist CD ROM via the 'CMC Remote Terminal' (refer to 46-20-01, Apex Electronic Checklist File – Load Procedures).

Table 1 MAU Module Software/Database Locations – Phase 1A Load 1.3

		SOFTWARE RESIDING ON CARD								
		APEX OPERATING SOFTWARE	**APM SETTINGS FILE** (NOTE 2)	**APM OPTIONS FILE** (NOTE 2)	**LDI LOADABLE DIAGNOSTIC INTERFACE**	**FMS PERFORM. DATABASE**	**NAV DATABASE (PURPLE)** (NOTE 1)	**CHARTS DATABASE (BLUE)**	**TERRAIN DATABASE (GREEN)**	**ELECT. CHECKLIST DATABASE**
MAU MODULE	NIC NO. 1	YES	YES	YES						
	NIC NO. 2	YES	YES	YES						
	AGM NO. 1	YES			YES		YES	YES	YES	
	AGM NO. 2	YES					YES	YES	YES	
	AIOP a	YES				YES	YES			
	AIOP b	YES				YES	YES			YES
	DGIO	YES								
	CSIO	YES								
	NO. 1 CONFIG. MODULE		YES	YES						
	NO. 2 CONFIG. MODULE		YES	YES						

Note

1 The NAV data base (purple) holds two individual files, CRH (Cities, Roads, Highways), and the INAV-16M, airspace communication and navigational data. Both files have to be individually loaded, using the CMC Remote Terminal Software (Maintenance Laptop).

2 The APM Option and the APM settings data is stored in the Aircraft Personally Modules, located within the backshells of the NIC No. 1 and NIC No. 2 cards.

Table 2 MAU Module Software/Database Locations – Phase 2A Load 4.4

		SOFTWARE RESIDING ON CARD								
		APEX OPERATING SOFTWARE	**APM SETTINGS FILE** (NOTE 2)	**APM OPTIONS FILE** (NOTE 2)	**LDI LOADABLE DIAGNOSTIC INTERFACE**	**FMS PERFORM. DATABASE**	**NAV DATABASE (PURPLE)** (NOTE 1)	**CHARTS DATABASE (BLUE)**	**TERRAIN DATABASE (GREEN)**	**ELECT. CHECKLIST DATABASE**
MAU MODULE	NIC NO. 1	YES	YES	YES						
	NIC NO. 2	YES	YES	YES						
	AGM NO. 1	YES					YES	YES	YES	
	AGM NO. 2	YES					YES	YES	YES	
	AIOP a	YES			YES	YES	YES			
	AIOP b	YES				YES	YES			YES
	DGIO	YES								
	CSIO	YES								
	NO. 1 CONFIG. MODULE		YES	YES						
	NO. 2 CONFIG. MODULE		YES	YES						

Note

1 The NAV data base (purple) holds two individual files, CRH (Cities, Roads, Highways), and the INAV-16M, airspace communication and navigational data. Both files have to be individually loaded, using the CMC Remote Terminal Software (Maintenance Laptop).

2 The APM Option and the APM settings data is stored in the Aircraft Personally Modules, located within the backshells of the NIC No. 1 and NIC No. 2 cards.

F. General Laptop CMC Remote Terminal Setup

1) The following procedure is required to load:

a) APEX MAU Operating Software

b) Settings File (refer to 45-00-01, Apex Options and Settings File – Load Procedures)

c) Options File (refer to refer to 45-00-01, Apex Options and Settings File – Load Procedures)

d) MAU LDI (refer to 45-00-05, Loadable Diagnostic Interface File – Load Procedures)

e) FMS Performance Database (refer to 45-00-03, Apex FMS Aircraft Performance Database Files – Load Procedure)

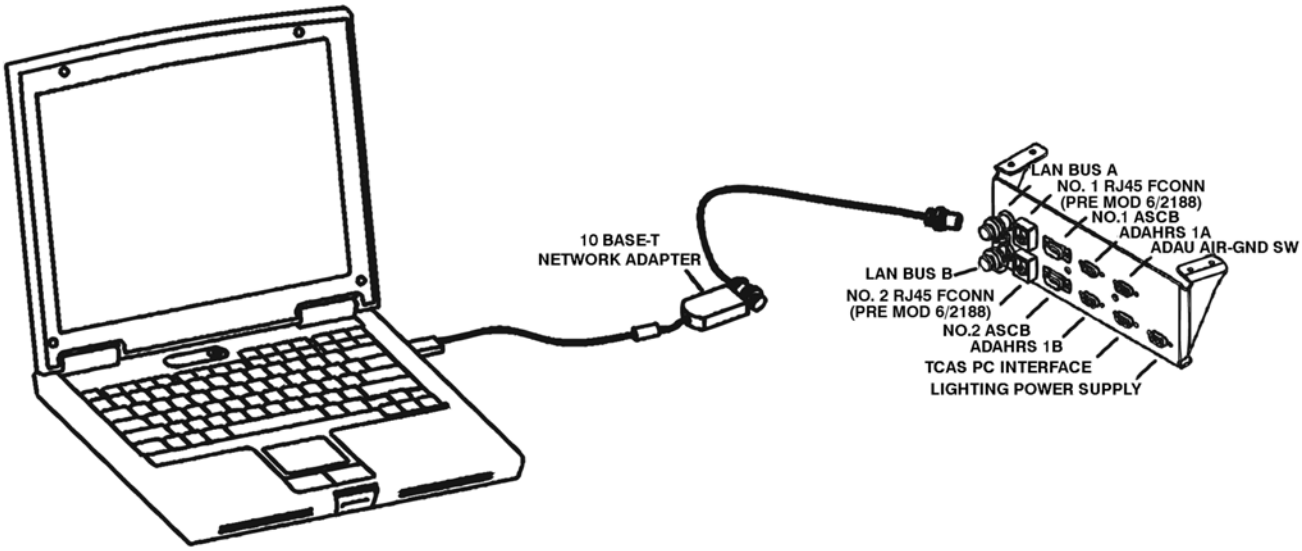
f) NAV Database (Purple) (refer to 45-00-04, Apex Database Files – Load Procedures)

g) CHART Database (Blue) (refer to 45-00-04, Apex Database Files – Load Procedures)

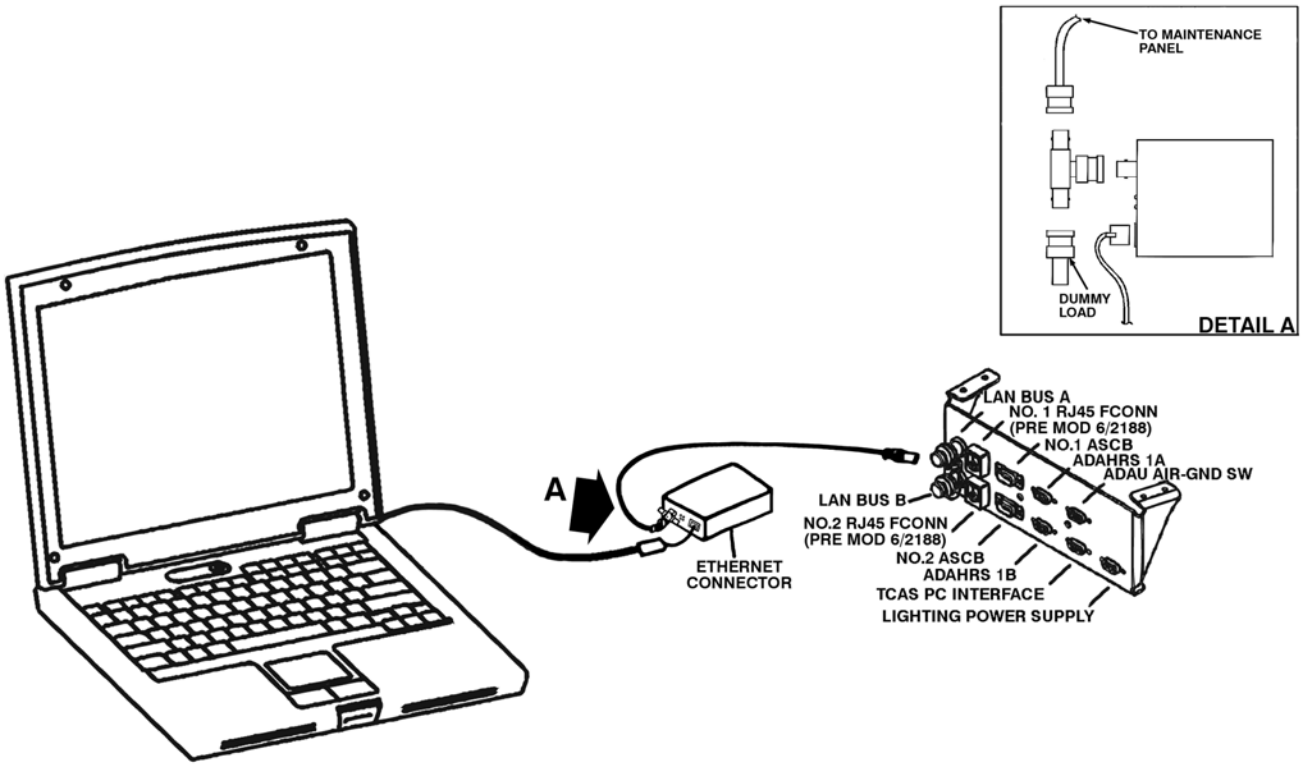
h) TERRAIN Database for MAU (Green) (refer to 45-00-04, Apex Database Files – Load Procedures)

2) Also required for:

Standard Maintenance features offered through the CMC remote tool.



Forward Maintenance Panel – PCMCIA Card
Figure 1



Forward Maintenance Panel – Ethernet to LAN Adapter Kit
Figure 2

G. Setup CMC Remote Terminal

1) This procedure requires the following:

a) Laptop computer with PCMCIA Card and 10BASE–T Network Adapter with LAN interface cable (refer to Figure 1), or Ethernet to LAN adaptor kit (refer to Figure 2).

b) Proprietary Honeywell CMC Remote Terminal Software RT 2.0 & DLS 5.0.2 or later applicable version (Pre Mod 6/2303); or RT 5.3 & DLS 7.3.1 or later applicable version (Post Mod 6/2303).

Note

- During loading, aircraft power must remain ON for the MAU system and for the Ethernet to LAN adapter.
- Ensure all MAU (4), ADAHRS (2), and MFD CTRL circuit breakers are closed. The MAU power must be on for a minimum of 15 minutes before beginning the loading procedure.
- Pitot/Static test sets may not be used during this procedure. Airspeed must be less than 40 knots and Rad Alt must be less than 10 feet.
- The MAU system must not be disturbed by other maintenance activities, which may affect the MAU system.
- APEX displays will be “X’d out” during this process. Do not attempt other activity on APEX during loading procedures.
- If loading has been disturbed in any way, there is a risk that loading will have to be repeated.
- This procedure requires the COM1 port of the laptop computer to be available. If problems are encountered disable other applications that may be using the COM1 port.

c) Gain access to the Avionics Maintenance Panel in the forward avionics area (refer to 12-00-00, Servicing – General Data).

d) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

e) Wait several minutes for the aircraft electrical and avionics systems to initialize.

f) Plug 10BASE–T Network Adaptor into laptop PCMCIA card (refer to Figure 1). Alternatively plug in Ethernet to LAN Adaptor Kit into Ethernet card on the laptop (refer to Figure 2).

g) On Maintenance Panel, on the Nose Avionics Rack, remove LAN terminator on LAN BUS A or LAN BUS B. (Refer to 12-00-00, Servicing – General Data).

Note Only remove one terminator at a time.

h) Connect LAN cable from adaptor to LAN BUS A connection on maintenance panel.

i) Insert the applicable CD or USB memory stick, which is holding the data to be loaded. Refer to applicable chapter for required files.

Note If the GIO card has been replaced, and the CMC Remote Terminal appears to not connect to the aircraft MAU, do the following to set the MAU into ground mode, to enable loading:

1 On the Maintenance Laptop open the Command Prompt:

WinXP:

- Click the ‘START’ button
- Click the ‘RUN’ button
- Type ‘cmd’
- Click ‘OK’

Win7:

- Click the ‘START’ button
- Type ‘RUN’ in the search field
- Type ‘cmd’
- Click ‘OK’

2 At the command prompt type each of the following commands followed by pushing the <ENTER> key:

- ftpshell 192.168.1.1
- quote site attrib 11 DEOS

3 Push <CTRL>-<C> on the keyboard

4 At the command prompt type each of the following commands followed by pushing the <ENTER> key:

- ftpshell 192.168.33.1
- quote site attrib 11 DEOS

5 Push <CTRL>-<C> on the keyboard

j) On laptop computer open the CMC Remote Terminal Program.

1 Open shortcut folder MAINT from desktop

2 Open MAU icon

3 Double click MAU CMC Remote Terminal

Note The shortcut may be placed at a different location on the laptop.

k) Select DATA LOADER from the CMC MAIN MENU. Select DATA LOADER again from the DATA LOADER MENU.

Note A 'Connection Failed' message will be witnessed prior to loading the LDI. CMC MAIN MENU will not be available, and pressing DATA LOADER will advance to the DLS INSTALLATION FUNCTION. It is possible for a 'Connection Failed' message to be displayed with the LDI already loaded. If the connection has not been established within 2 minutes, check cables for correct connection. Check LED's on network converter. LED's flash when operating correctly. Ensure Ethernet to LAN converter has a power source.

l) Once PRE-LOADING DMU/CDROM DRIVES show's 'DONE', select FULL LOAD.

H. Loading File/Data:

1) Navigate to the data file to be loaded:

a) Arrow down and select 'AVAILABLE DRIVES'.

Note To navigate with the arrow keys, the mouse cursor must be moved over the selection window. The window is highlighted in green when active.

b) Open up applicable folders by pushing the <ENTER> button on the displayed folder.

c) Select the file which is prefixed by [DR-PN] for the current DATA to be loaded. To select, highlight file and press <ENTER>. Selected file will be highlighted in green.

Note Non - [DR-PN] files are not displayed. The program will respond with an error if the connection to the MAU is lost or if the file is not for the DHC-6 Series 400 Twin Otter.

d) Click 'SELECT FILE' in the menu. 'CONFIGURATION CHECK IN PROGRESS' will be displayed. Configuration check may take several minutes.

Note When loading the options file, the following additional steps are required before configuration check is performed:

1 When 'SELECT FILE' is selected, the Options File Licensing Agreement is displayed.

2 Review agreement using the arrow keys to scroll to bottom.

3 Highlight 'I ACCEPT' and press <ENTER> to select, when review is complete.

4 Select 'ENTER CODE' in the menu.

5 Enter code from CD into the AUTHORIZATION CODE window.

6 Select 'ENTER' in the menu. Confirm red password accepted message. 'CONFIGURATION CHECK IN PROGRESS' will be displayed.

e) Wait until 'CONFIGURATION CHECK COMPLETE' is displayed. Estimated load times will be displayed.

f) Click ‘START LOAD’ in the menu.

g) Wait until ‘LOADING SEQUENCE COMPLETE’ is displayed. The loading process can take several minutes depending on the file to be loaded. Refer to applicable chapter for file loading times.

Note If ‘Ping/FTP Communication Fail” error message is displayed during load, cycle aircraft power, close/reopen the CMC application on the maintenance laptop, and repeat load procedure. If message persists, the file may be corrupt. Obtain a new copy of the file, if able.

h) On the Remote Terminal, verify installation completed successfully on each listed card. For files affecting a large number of cards, it may be necessary to scroll down, using the arrow keys, to view the entire list.

Note If the installation failed on any card, the process must be repeated. The installation will detect the status of each card, and only repeat the process on required cards.

i) Click on ‘MAIN MENU’ button, if more data files are to be loaded. Insert applicable media CD or USB stick. Repeat Para 1.H.1).

I. Closing CMC Remote Terminal

1) Proceed by the following:

a) Close CMC Remote Terminal Program.

b) Disconnect adaptor LAN cable from the maintenance panel LAN BUS A.

c) Reconnect LAN BUS A terminator on the maintenance panel.

d) Remove aircraft power from APEX for a minimum of one minute.

e) Reapply power to APEX.

f) Verify no APEX errors are displayed following initialization.

g) Via the aircraft cockpit displays and controls, the APEX SCMS/DL page may be used to verify APEX software, configuration settings and databus status.

Note Refer to the chapters applicable to the data that was loaded for verification procedures.

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